PHYSICAL CHEMISTRY

GUIDED REVISION PLAN_SCORE ADVANCED [SOLUTION]

Topic : MOLE CONCEPT & REDOX (Question Bank # 05)

SINGLE CORRECT

1. (B)

Sol. $Na_2CO_3 \cdot NaHCO_3 \cdot 2H_2O \xrightarrow{0.05\,M\,HCl}$

10 ml, 0.1 M

[Acid base titration - case of non-redox reaction]

(I) At phenol phthalein end point : Only Na_2CO_3 is converted in $NaHCO_3$.

Balanced chemical equation :

$Na_2CO_2 \cdot NaHCO_3 \cdot 2H_2O + HCl \longrightarrow 2NaHCO_3 + 2H_2O + NaCl$

10ml, 0.1M 0.05M

x ml

→ Since molar ratio is 1 : 1

$\therefore\ 10 \times 0.1 = x \times 0.05 \Rightarrow \boxed{x = 20\,ml}$

(II) Methyl orange end point : All the $NaHCO_3$ is converted in H_2CO_3

$Na_2CO_3 \cdot NaHCO_3 + 3HCl \longrightarrow 3H_2CO_3 + 3NaCl$

0.1M, 10ml 0.05M

y ml

→ Since molar ratio is 1 : 3

$\therefore\ \dfrac{10 \times 0.1}{0.05 \times y} = \dfrac{1}{3} = \boxed{y = 60\,ml}$

→ Hence $y - x = 60 - 20 = 40$ ml

2. (B)

Sol. (i) $H_2\underline{\overset{+3}{C_2}}O_4 + \underline{\overset{+7}{Mn}}O_4^- \longrightarrow \underline{\overset{+4}{CO_2}} + Mn^{+2}$

$n = 2$ $n = 5$

100 ml

(ii) $\underline{\overset{+7}{Mn}}O_4^- + I^- \longrightarrow Mn^{+2} + I_2$

10 ml 50 ml

n = 5 1N

From equation (ii) : m. eq. of MnO_4^- = m. eq. of I^-

$\Rightarrow M \times 5 \times 10 = 1 \times 50$

$\Rightarrow \boxed{M_{MnO_4^-} = 1M}$

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From equation (i) :

$$\text{m. eq. of } H_2C_2O_4 = \text{m. eq. of } MnO_4^-$$

$$\Rightarrow x \times 2 = 1 \times 100 \times 5$$

$$\Rightarrow x = 250 \text{ m. mol} = \frac{1}{4} \text{ mol}$$

$$\therefore W_{H_2C_2O_4} = \frac{1}{4} \times 90\,g$$

$$= 22.5\ g$$

3. (D)

Sol. $\boxed{2M\,AlCl_3 + 3m\ SnCl_4}$

Sincle $[\text{Molarity}]_{Cl^-} = [\text{Molality}]_{Cl^-}$

$$\Rightarrow \frac{n_{Cl^-}}{V_{solution}} \text{in L} = \frac{n_{Cl^-}}{W_{solvent}} \text{in kg}$$

$$\Rightarrow \boxed{V_{solution} \text{ in L} = W_{solvent} \text{in kg}}$$

$\rightarrow$ Consider 1 L solution [that means 1 kg solvent]

therefore : $n_{AlCl_4} = 2\,mol \Rightarrow W_{AlCl_3} = 267\,g$

& $n_{SnCl_4} = 3mol \Rightarrow W_{SnCl_4} = 783\,g$

Therefore $W_{solution} = W_{AlCl_3} + W_{SnCl_4} + W_{solvent}$

$= 267 + 783 + 1000$

$= 2050\ g$

Now:

$$\text{density of Soln} = \frac{W_{solution}}{V_{solution}} = \frac{2050\,g}{1000\,ml} = 2.05\,g/ml$$

4. (B)

Sol. $\boxed{\begin{matrix}150\,ml \\ I_2 : 'x'\end{matrix}}$ M

Part-I : $I_2 + Na_2\overset{+2}{\underline{S_2}}O_3 \longrightarrow NaI + Na_2\overset{+2.5}{\underline{S_4}}O_6$

$n = 2 \quad n = 1$

'x' M 15 ml, 0.4 m

V ml

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→ Acc. to law of equivalence :

m. eq. of I_2 = m. eq. of $Na_2S_2O_3$

$\Rightarrow V \times x \times 2 = 15 \times 0.4 \times 1$

$\Rightarrow \boxed{V \times x = 3}$ ....(1)

Part-II :

(A) $3I_2 + 6\overset{\ominus}{O}H \longrightarrow 5I^- + IO_3^-$ [Balanced]

(150-V)ml (excess)

'x' M 100 ml, 0.3 M

= 30 m. mol

= 24 m. mol (consumed)

[Remaining]

(B) $2NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O$

6m mol 10 ml

0.3 M

= 3 m. mol

from equation (A) : $\frac{[\text{m.mol}]_{I_2}}{[\text{m.mol}]_{\overset{\ominus}{O}H}} = \frac{1}{2}$

$\Rightarrow (150 - V) \times x = \frac{1}{2} \times 24 = 12$

$\Rightarrow 150x - Vx = 12 \Rightarrow 150x = 15$

$\Rightarrow \boxed{x = 0.1M}$

5. (A)

Sol. $\boxed{458\,g \text{ Sample } Mn_3O_4}$

(A) (I) $Mn_3O_4 \xrightarrow{F^-} 3Mn^{+2}$

1 mol 3 mol

= 229 g

(II) $Mn^{2+} + K\underset{}{\overset{+7}{Mn}}O_4 \longrightarrow Mn^{+3}$

n = 1 n = 4

3 eq 3 L

3 mol $\frac{1}{4}M$

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(B) $K\overset{+7}{\underline{Mn}}O_4 + H_2C_2O_4 \longrightarrow Mn^{+2} + CO_2$

1.25 N

n = 5

$\text{Molarity} = \frac{1}{4}M$

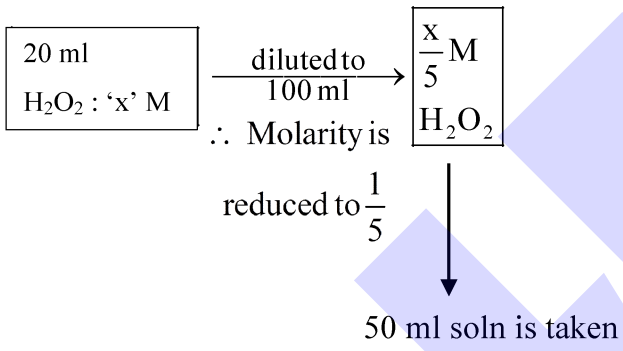
$\rightarrow$ therefore :

$$\%\text{ purity in the mixture} = \frac{W_{Mn_3O_4}}{W_{Sample}} \times 100$$

$$= \frac{229}{458} \times 100$$

6. **(C)**

Sol.



$\rightarrow$ $H_2\overset{-1}{\underline{O}}_2 + K\overset{+7}{\underline{Mn}}O_4 \xrightarrow{\overset{\oplus}{H}} \overset{0}{\underline{O}}_2 + Mn^{+2}$

n = 2 n = 5

$\frac{x}{5}M$ 0.02 M

50 ml 50 ml

$\rightarrow$ Acc. to law of equivalence

$$\frac{x}{5} \times 50 \times 2 = 50 \times 0.02 \times 5$$

$$\Rightarrow \boxed{x = \frac{1}{4}M}$$

$\rightarrow$ Now volume strength of H_2O_2 solution

(at 1 atm, 273 k)

= 11.2 × Molarity = 2.8 V

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7. (B)

Sol. $\boxed{NaHC_2O_4\,(x\text{ mol}) + KHC_2O_4\cdot H_2C_2O_4\,(y\text{ mol})}$

Case-I: Reaction with $KMnO_4$: [Redox reaction]

[Reacting species is $C_2O_4^{-2}$]

$$\overset{+3}{\underline{C}}_2\,O_4^{-2} \;+\; K\overset{+7}{\underline{M}}nO_4 \longrightarrow \overset{+4}{\underline{C}}O_2 + Mn^{+2}$$

$(x+2y)$ mol $\qquad n = 5$

$n = 2 \qquad V,\ 0.2\ N$

$\rightarrow$ Law of equivalence : $(x + 2y) \times 2 = 0.2 \times V$

$\Rightarrow \boxed{V = 10(x+2y)} \qquad ...(i)$

Case-II: Reaction with 0.12 N NaOH : (Non-redox reaction)

[Acid-base reaction]

$$\overset{\oplus}{H} \;+\; \overset{\ominus}{O}H \longrightarrow H_2O$$

$(x+3y)$ mol $\qquad 0.12\ N$

$n = 1 \qquad V$

$\Rightarrow (x+3y) \times 1 = 0.12 \times V$

$\Rightarrow \boxed{V = \frac{100}{12}\times[x+3y]} \qquad ....(ii)$

(i) / (ii)

$\boxed{\frac{x}{y} = \frac{3}{1}}$

8. (D)

Sol. Kjeldahl method :

$$\underset{1.4\,g}{\boxed{O.C.}} \xrightarrow{\text{evolved } NH_3} NH_3$$

(I) $2NH_3 \;+\; H_2SO_4 \longrightarrow (NH_4)_2\,SO_4$

$10\text{m. mol} \qquad 60\text{ml}, \frac{M}{10}$

$= 6$ m. mol [Initial]

$= 5$ m. mol [Consumed in the reaction].

Remaining

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(II) $H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2H_2O$

1 m. mol 20 ml

$\frac{M}{10}$

$= 2$ m. mol

therefore m. mol of NH_3 produced $= 10$

$\Rightarrow$ Since m. mol of N = m. mol of NH_3

$\Rightarrow$ m. mol of N = 10

$\Rightarrow$ mol of $N = \frac{10}{1000}$

$\Rightarrow W_N = \frac{1}{100} \times 14\,g$

$$\text{therefore } \%\,N = \frac{W_N}{W_{O.C.}} \times 100 = \frac{\frac{14}{100}}{1.4} \times 100$$

$= 10\%$

9. (A)

Sol. $\boxed{FeO\,(a\,mol) + Fe_2O_3\,(b\,mol)}$

Case-I : $KMnO_4$ reacts only with FeO, since it is an oxidizing agent.

$\rightarrow K\overset{+7}{\underline{M}}nO_4 + \overset{+2}{\underline{Fe}}O \longrightarrow Fe^{+3} + Mn^{+2}$

$\frac{2}{5}$ M, $\boxed{n = 5}$ a mol $n = 1$

0.1 L $n = 1$

* $\boxed{\text{Eq. of FeO} = \text{Eq. of } KMnO_4}$ = eq. of Fe^{+3}

$\Rightarrow a \times 1 = \frac{2}{5} \times 0.1 \times 5 \Rightarrow \boxed{a = \frac{2}{10}\,mol}$

Case-II: Now, Zn reduced all the Fe^{+3} in Fe^{+2}

$\rightarrow$ Total amount of Fe^{+3} in mix $= (a + 2b)$ mol

{'a' mol is obtained from the oxidation of Fe^{+2} and '2b' mol of Fe^{+3} from Fe_2O_3}

$Zn + Fe^{+3} \longrightarrow Zn^{+2} + Fe^{+2}$

(a+2b) mol –

– (a+2b) mol

[POAC]

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Case-III: $Fe^{+2} + \overset{+6}{K_2Cr_2O_7} \longrightarrow Fe^{+3} + Cr^{+3}$

$n = 1 \qquad n = 6$

$(a+2b)\text{mol} \quad \frac{2}{15}M_1\,1L$

$$= (a+2b)\times 1 = \frac{2}{15}\times 1\times 6 \Rightarrow b = \frac{3}{10}\text{mol}$$

To calculate $\frac{a}{b} \Rightarrow \frac{a}{b} = \frac{\frac{2}{10}}{\frac{3}{10}} = \frac{2}{3}$

MULTIPLE CORRECT

10. (B,C)

Sol. Let volume of $NH_{3(g)} = V$ ml

Dissociation of $NH_{3(g)}$:

$$2NH_{3(g)} \longrightarrow N_{2(g)} + 3H_{2(g)}$$

$t = 0$	V	–	–
t	$V - V_1$	$\frac{V_1}{2}$	$\frac{3V_1}{2}$

→ After dissociation of NH_3, volume becomes 90 ml. [NH_3 is not completely dissociated, since after reaction with H_2SO_4 volume is converted in 80 ml. H_2SO_4 reacts with NH_3 completely to get $(NH_4)_2SO_4$

→ therefore after reaction:

* $V_{N_2} + V_{H_2} = 80$

$$\Rightarrow \frac{V_1}{2} + \frac{3V_1}{2} = 80 \Rightarrow \boxed{V_1 = 40\,\text{ml}}$$

→ and remaining vol. of NH_3 :

$$V_{NH_3} = V - V_1 = 10$$

$$\Rightarrow \boxed{V = 50\,\text{ml}}$$

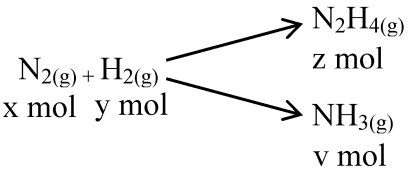
→ that mean 80% NH_3 is dissociated in N_2 and H_2.

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11. (A,B,D)

Sol.



Chemical reactions are

Step-I) $N_{2(g)} + 2H_{2(g)} \longrightarrow N_2H_{4(g)}$

Step-II) $N_2H_{4(g)} + H_{2(g)} \longrightarrow 2NH_{3(g)}$

→ Check first option :

(A) (2, 2, 1, 0)

(I) $N_{2(g)} + \overset{(LR)}{2H_{2(g)}} \longrightarrow N_2H_{4(g)}$

2mol	2mol	–
1mol	–	1mol

Next step is possible only if H_2 is in excess, but since it is LR, therefore $n_{NH_3} = 0$

First option is correct.

(B) (3, 8, 1, 4)

(I) $\overset{(LR)}{N_{2(g)}} + \overset{(Excess)}{2H_{2(g)}} \longrightarrow N_2H_{4(g)}$

3mol	8mol	–
–	2 mol	3 mol

II reaction is possible.

(II) $N_2H_{4(g)} + \overset{(L.R)}{H_{2(g)}} \longrightarrow 2NH_{3(g)}$

3 mol	2 mol	–
1 mol	–	4mol

Final cond. N_2H_4 : 1 mol

NH_3 : 4 mol

→ Second option is also correct.

→ Similarly check other options.

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12. (A)

Sol. * 1 L water sample is taken.

* Heavily polluted means, concentration of dissolved oxygen (O_2) in water is less than 6.4 mg/L

* Mass of dissolved O_2 in 1L sample

$$= 6.4\frac{mg}{L} \times 1L$$

$$= 6.4\ mg$$

* m. mol of dissolved $O_2 = \frac{6.4}{32} = 0.2$

→ from the equation :

(I) $Mn(OH)_2 + \frac{1}{2}O_2 + H_2O \longrightarrow 2Mn(OH)_{3(s)}$

0.2 m.mol –

– 0.8 m. mol

(II) $2Mn(OH)_3 + 2KI + 3H_2SO_4 \longrightarrow 2MnSO_4 + k_2SO_4 + I_2 + 6H_2O$

0.8 m. mol –

– 0.4 m. mol

(III) $I_2 + 2Na_2S_2O_3 \longrightarrow 2NaI + Na_2S_4O_6$

0.4 m. mol –

– 0.8 m. mol

→ therefore m. mol of $Na_2S_2O_3$ required = 0.8

⇒ $M \times V = 0.8$

⇒ $0.01 \times V = 0.8$

⇒ $\boxed{V = 80\,ml}$

13. (C)

Sol. 1L water sample is taken :

→ concentration of O_2 required for optimum fit for aquatic life = 9.6 mg/L

→ ∴ m.mol of $O_2 = \frac{9.6}{32}\frac{m.mol}{L} \times 1L$

$= 0.3$ m. mol

→ from the equation :

(I) $2Mn(OH)_2 + \frac{1}{2}O_2 + H_2O \longrightarrow 2Mn(OH)_{3(s)}$

0.3 m. mol –

– 1.2 m. mol

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(II) $2Mn(OH)_{3(s)} + 2KI + 3H_2SO_4 \longrightarrow 2MnSO_4 + K_2SO_4 + I_2 + 6H_2O$

1.2 m. mol –

– 0.6 m. mol

therefore minimum m. mol of $Na_2S_2O_3$ required = 1.2

$\Rightarrow$ M × V = 1.2

$\Rightarrow$ 0.1 × V = 1.2

$\Rightarrow$ $\boxed{V = 12\,ml}$

14. (B)

Sol. (B) is the correct statement.

(A) Incorrect : more dissolved O_2 in water requires more moles of $Na_2S_2O_3$

(C) Incorrect : $n_{Mn(OH)_2}$ and $n_{Mn(OH)_3}$ will be same (according to reaction stoichiometry)

(D) Incorrect : 1 mol of dissolved O_2 will require 4 mol of $Na_2S_2O_3$ for estimation.

SECTION-II [INTEGER TYPE]

1. (4)

Sol. $[\text{density}]_{(C_2H_5)_4Pb} = 6.48\,g/ml,\ vol = 2ml$

$\therefore$ Mass of $(C_2H_5)_4Pb$ reqd. for 1L gasoline

= v × d = 2 ml × 6.48 g/ml

= 2 × 6.48 g

$\therefore$ for 25 L gasoline = 2 × 6.48 × 25 g

From the given equation:

$4\,C_2H_5Cl + 4\,Na - Pb \longrightarrow (C_2H_5)_4Pb + 4\,NaCl + 3\,Pb$

– $\frac{2\times6.48\times25}{324}$ mol

4 mol = 1 mol

Ans. 4 mol C_2H_5Cl

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2. **(1)**

Sol. $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24\,H_2O$

Given: $W_0 = 5000\ g$

$$n_0 = \frac{5000}{16}\text{mol}$$

Since molar ratio of 'S' and 'O' is $\frac{1}{10}$

therefore: $\frac{n_S}{n_O} = \frac{1}{10}$

$$\Rightarrow \quad n_S = \frac{1}{10} \times n_O$$

$$= \frac{1}{10} \times \frac{5000}{16}\text{mol}$$

therefore: $W_S = \frac{5000}{10 \times 16} \times 32g$

$= 1\ kg$

3. **(5)**

Sol. $\underset{n_2}{Br_2} + OH^{\ominus} \longrightarrow Br^- + \text{Oxidized product}$ ($+2e^-$, $n_1 = 2$)

→ D is proportionation reaction :

* n - factor of $Br_2 = \frac{n_1 n_2}{n_1 + n_2}$

→ Given eq. wt. of $Br_2 = 96$

$$\Rightarrow \frac{M_{Br_2}}{n - \text{factor}} = 96$$

$$\Rightarrow n - \text{factor} = \frac{160}{96} = \frac{5}{3}$$

$$\Rightarrow \frac{n_1 \times n_2}{n_1 + n_2} = \frac{5}{3} \qquad \{ n_1 = 2$$

$$\Rightarrow \boxed{n_2 = 10}$$

To loose 10 mol of e^- & by 1 mol Br_2. Oxidation No. of 'Br' in the oxixized product should be '+5'.

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SECTION-IV [MATRIX MATCH]

1. (A-Q,R; B-P,S; C-Q,R)

Sol. Two separate solution :

$KMnO_4 \xrightarrow[\text{Medium}]{\text{Neutral}} Mn^{+4}$, $n = 3$

$O_2 \longrightarrow O^{-2}$, $n = 4$

*** First experiment :**

(I) $KMnO_4 + \overset{+2}{\underline{Fe}}\overset{+3}{\underline{Cr_2}}O_4 \longrightarrow Mn^{+4} + \overset{+3}{\underline{Fe_2}}O_3 + \underline{K_2}\overset{+6}{Cr}O_4$

$n = 3$

$\frac{1}{20}$M, 20 ml $\qquad n = 7$

$\rightarrow$ m. eq. of $KMnO_4$ = m. eq. of $FeCr_2O_4$

$\Rightarrow \frac{1}{20} \times 20 \times 3 = \text{m.mol} \times 7$

$\Rightarrow (\text{m.mol})_{FeCr_2O_4} = \frac{3}{7}$

(II) $O_2 + FeCr_2O_4 \longrightarrow O^{-2} + Fe_2O_3 + K_2CrO_4$

$n = 4 \qquad n = 7$

$\frac{1}{10}$M, 10 ml

$\rightarrow$ m.eq. of O_2 = m.eq. of $FeCr_2O_4$

$\Rightarrow \frac{1}{10} \times 10 \times 4 = \text{m.mol} \times 7$

$\Rightarrow [\text{m.mol}]_{FeCr_2O_4} = \frac{4}{7}$

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Experiment-II :

* n-factor of $CuFeS_2$:

$$CuFeS_2 + 4H_2O \longrightarrow Cu^{+} + Fe^{+2} + 2SO_2 + 8\overset{\oplus}{H}$$

$$\boxed{n = 11} \qquad +11e^{-}$$

(I) m. eq. of $KMnO_4$ = m. eq. of $CuFeS_2$

$$\Rightarrow \frac{1}{20} \times 20 \times 3 = m.mol \times 11$$

$$\Rightarrow (m.mol)_{CuFeS_2} = \frac{3}{11}$$

(II) m. eq. of O_2 = m. eq. of $CuFeS_2$

$$\Rightarrow [m.mol]_{CuFeS_2} = \frac{4}{11}$$

Experiment-III :

* n - factor of FeS :

$$FeS + 2H_2O \longrightarrow Fe^{+3} + SO_2 + 4\overset{\oplus}{H} + 7e^{-}$$

$$\boxed{n = 7}$$

(I) $\boxed{m.eq.of\ KMnO_4 = m.eq.of\ FeS}$

(II) $\boxed{m.eq.of\ O_2 = m.\ eq.of\ FeS}$

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